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Time taken 6 mins 24 secs

Grade 10.00 out of 10.00 (100%)

Question 1

Correct

Mark 1.00 out of 1.00

Which node is **purier**?

- ☐ a. Node B: (6+, 4-)
- ☐ b. Cannot be determined
- ☐ c. Both equally pure
- ☒ d. Node A: (9+, 1-) ✓

Your answer is correct.

The correct answer is: Node A: (9+, 1-)

Question 2

Correct

Mark 1.00 out of 1.00

Which type of split is used by the **CART algorithm**?

- ☐ a. Ordinal split
- ☐ b. Multi-way split only
- ☐ c. Random split
- ☒ d. Binary split only ✓

Your answer is correct.

The correct answer is: Binary split only

Question 3

Correct

Mark 1.00 out of 1.00

Sampling in Bagging is done:

- ☒ a. With replacement ✓
- ☐ b. Using weighted data
- ☐ c. Without replacement
- ☐ d. Sequentially

Your answer is correct.

The correct answer is: With replacement

Question 4

Correct

Mark 1.00 out of 1.00

Which feature type requires threshold selection during splitting?

- ☐ a. Ordinal
- ☐ b. Binary
- ☒ c. Continuous ✓
- ☐ d. Nominal

Your answer is correct.

The correct answer is: Continuous

Question 5

Correct

Mark 1.00 out of 1.00

Based on the Adaboost, consider the following:

$$\alpha_1 = 0.6, h_1(x) = 1$$

$$\alpha_2 = 0.3, h_2(x) = -1$$

Compute the final prediction:

- ☐ a. None of these answers
- ☐ b. 0
- ☐ c. -1
- ☒ d. +1 ✓

Your answer is correct.

The correct answer is: +1

Question 6

Correct

Mark 1.00 out of 1.00

What does a **Gini Index of 0** indicate?

- ☐ a. Maximum uncertainty
- ☐ b. Perfect misclassification
- ☐ c. Equal class probabilities
- ☒ d. A completely pure node ✓

Your answer is correct.

The correct answer is: A completely pure node

Question 7

Correct

Mark 1.00 out of 1.00

Which two sources of randomness are used in Random Forest?

- ☐ a. Random depth and random pruning
- ☒ b. Random features and random subsets of data ✓
- ☐ c. Random labels and random targets
- ☐ d. Random loss functions and gradients

Your answer is correct.

The correct answer is: Random features and random subsets of data

Question 8

Correct

Mark 1.00 out of 1.00

What happens to misclassified samples in AdaBoost?

- ☒ a. Their weights increase ✓
- ☐ b. They are removed
- ☐ c. They are ignored
- ☐ d. Their weights decrease

Your answer is correct.

The correct answer is: Their weights increase

Question 9

Correct

Mark 1.00 out of 1.00

In Decision Trees, **pruning** is primarily used to:

- ☐ a. Reduce tree depth to zero
- ☒ b. Improve generalization by reducing overfitting ✓
- ☐ c. Increase training accuracy
- ☐ d. Increase information gain

Your answer is correct.

The correct answer is: Improve generalization by reducing overfitting

Question 10

Correct

Mark 1.00 out of 1.00

Gradient Boosting differs from AdaBoost because it:

- ☐ a. Trains models in parallel
- ☐ b. Uses voting
- ☐ c. Uses bootstrapping only
- ☒ d. Fits residual errors using gradients ✓

Your answer is correct.

The correct answer is: Fits residual errors using gradients